COURSE MATERIAL	
SUBJECT	BASIC CIVIL & MECHANICAL ENGINEERING (ME23AES101)
UNIT	1
COURSE	B.TECH
DEPARTMENT	ME
SEMESTER	1-1
PREPARED BY (Faculty Name/s)	Mr.K.Harshavardhan Reddy Assistant Professor
Version	V-1
PREPARED / REVISED DATE	19-03-2024

1. Course Objectives

The objectives of the course are to make the students learn about:

1. Get familiarized with the scope and importance of Mechanical Engineering in different sectors and industries.
2. Explain different engineering materials and different manufacturing processes.
3. Provide an overview of different thermal and mechanical transmission systems and introduce basics of robotics and its applications.

2. Prerequisites

Students should have knowledge on

1. Engineering Physics

3. Syllabus

UNIT–III

Power plants – Working principle of Steam, Diesel, Hydro, Nuclear power plants.
Mechanical Power Transmission - Belt Drives, Chain, Rope drives, Gear Drives and their applications.
Introduction to Robotics - Joints & links, configurations, and applications of robotics.

4. Course outcomes

1. **Understand** the different manufacturing processes (L1).
2. **Explain** the basics of thermal engineering and its applications (L2).
3. **Describe** the working of different mechanical power transmission systems and power plants and describe the basics of robotics and its applications (L2).

5. Co-PO / PSO Mapping

Machine Tools	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3												3	
CO2	3						1						3	
CO3	3												3	

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Power plants – Working principle of Steam power plants.	T1, R1
2		Diesel, Hydro power plants.	T1, R1
3		Nuclear power plants.	T1, R1
4		Mechanical Power Transmission	T1, R1
5		Belt Drives, Chain Drives	T1, R2
6		Rope drives	T1, R1
7	2	Gear Drives and their applications.	T1, R1
8		Introduction to Robotics	T1, R1
9		Joints & links	T1, R1
10		configurations and applications of robotics.	T1, R1

7. Activity Based Learning

1. Mechanical Power Transmission

8. Lecture Notes

1.1 INTRODUCTION

Power plants are industrial facilities used to generate electricity by converting various forms of energy into electrical energy. These plants play a crucial role in meeting the energy demands of residential, commercial, and industrial sectors.

There are several types of power plants based on the source of energy used:

1. **Thermal Power Plant** – Uses fossil fuels (coal, oil, or gas) to heat water, producing steam that drives turbines.
2. **Hydroelectric Power Plant** – Utilizes the energy of falling or flowing water to rotate turbines.
3. **Nuclear Power Plant** – Generates heat through nuclear fission to produce steam for turbine rotation.
4. **Solar Power Plant** – Converts sunlight directly into electricity using photovoltaic cells or uses solar heat to produce steam.
5. **Wind Power Plant** – Uses the kinetic energy of wind to rotate wind turbines and generate electricity.

6. **Geothermal Power Plant** – Taps into the Earth's internal heat to produce steam for turbines.

1.1.1 Thermal Power Plant (Steam power plant):

A **Thermal Power Plant** is a facility that generates electricity by converting heat energy, usually obtained from the combustion of fossil fuels such as coal, oil, or natural gas, into electrical energy.

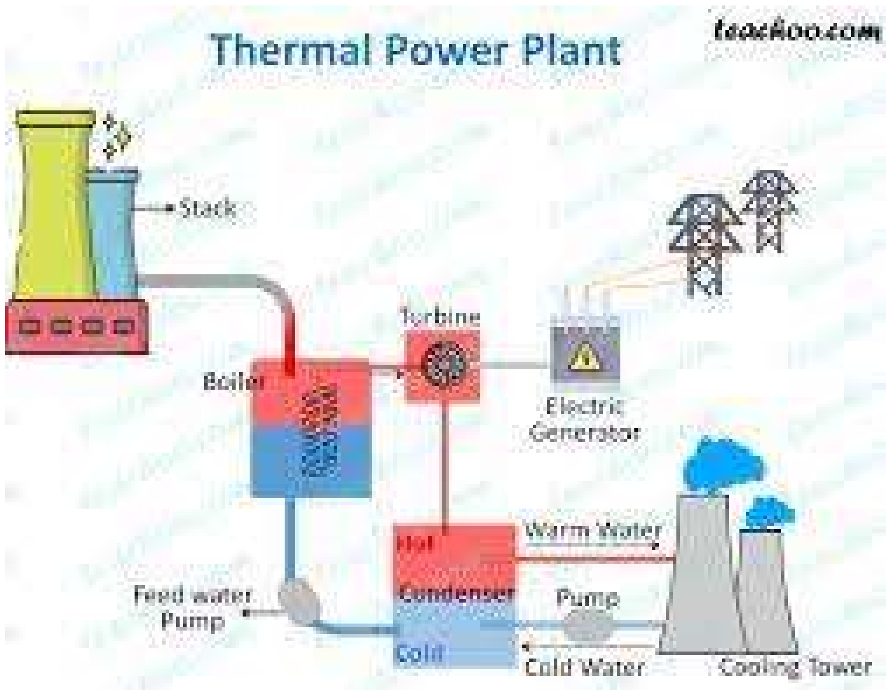

fig.3.1 Thermal Power Plant

Working Principle:

The basic principle behind a thermal power plant is the **Rankine cycle**. Heat energy is used to convert water into steam, which then rotates a turbine connected to a generator, producing electricity.

Main Components:

1. **Boiler/Furnace** – Burns fuel to produce heat.
2. **Economizer** – Heats feed water using flue gases.
3. **Steam Turbine** – Converts steam's thermal energy into mechanical energy.
4. **Generator** – Converts mechanical energy into electrical energy.
5. **Condenser** – Cools the exhaust steam from the turbine back into water.

6. **Cooling Tower** – Dissipates heat into the atmosphere.
7. **Chimney** – Releases exhaust gases into the air.

Advantages:

- Can be built anywhere with fuel and water availability.
- Continuous and stable power generation.
- Less initial cost compared to hydro or nuclear plants.

Disadvantages:

- Emits greenhouse gases (CO_2, SO_2, NO_2).
- High operating and maintenance cost.
- Non-renewable fuel usage leads to depletion of natural resources.

1.1.2 Nuclear Power Plant:

A Nuclear Power Plant is a facility that generates electricity using the heat produced by nuclear fission. In this process, the nucleus of a heavy atom (typically uranium-235 or plutonium-239) splits into smaller parts, releasing a large amount of heat energy, which is then used to produce steam that drives turbines and generates electricity.

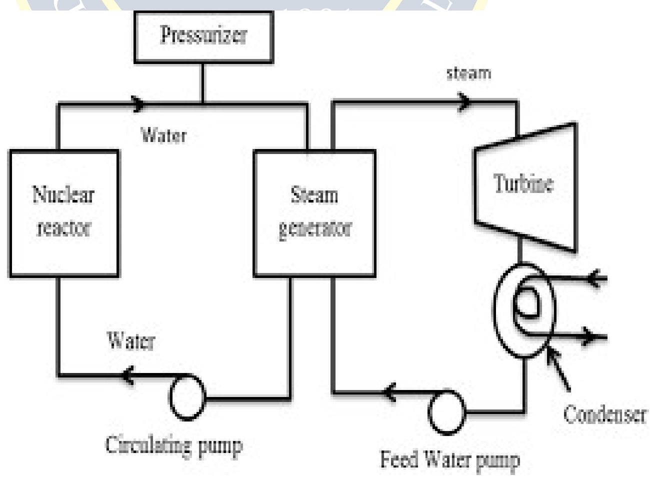

Fig.3.2 Nuclear Power Plant

Working Principle:

Nuclear power plants work on the Rankine cycle, similar to thermal plants, but the heat source is nuclear fission instead of fossil fuel combustion.

Main Components:

1. **Nuclear Reactor** – The core unit where fission occurs, generating heat.
2. **Fuel Rods** – Contain nuclear fuel (usually uranium).
3. **Control Rods** – Absorb neutrons to control the fission rate.
4. **Moderator** – Slows down neutrons to sustain chain reactions (commonly water or graphite).
5. **Coolant** – Transfers heat from the reactor to the steam generator (e.g., water, gas, or liquid metal).
6. **Steam Generator** – Converts heat from coolant into steam.
7. **Turbine and Generator** – Steam rotates the turbine, which drives the generator to produce electricity.
8. **Condenser** – Converts exhaust steam back into water.
9. **Containment Structure** – A thick, sealed building that encloses the reactor to prevent radiation leaks.

Advantages:

- Produces large amounts of electricity with very low green house gas emissions.
- Requires less fuel compared to fossil fuel plants.
- Efficient and reliable power generation.

Disadvantages:

- High initial construction and maintenance costs.
- Radioactive waste disposal is a major concern.
- Risk of radiation leaks or accidents (e.g., Chernobyl, Fukushima).
- Requires strict safety and regulatory measures.

1.1.3 Hydro Power Plant:

A Hydro Power Plant (or Hydroelectric Power Plant) is a facility that generates electricity by harnessing the kinetic and potential energy of flowing or falling water. It is one of the most widely used renewable energy sources.

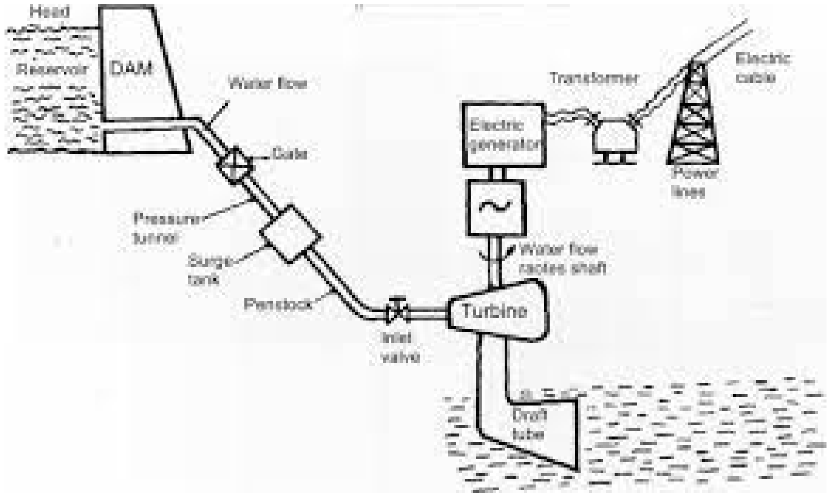

fig.3.3 Hydro Power Plant

Working Principle:

The hydro power plant operates on the principle of converting potential energy of stored water into mechanical energy by allowing it to flow through turbines, which then drive generators to produce electricity.

Main Components:

1. Dam – Stores water at a height, creating potential energy.
2. Reservoir – A water storage body behind the dam.
3. Penstock – Large pipes that carry water from the reservoir to the turbines.
4. Turbine – Rotated by water flow, converting hydraulic energy into mechanical energy.
5. Generator – Converts mechanical energy from the turbine into electrical energy.
6. Tailrace – Carries water away from the turbine back to the river.

Advantages:

1. Renewable and clean energy source.
2. Low operating cost after construction.
3. Helps in irrigation and flood control.
4. Long lifespan and high efficiency.

Disadvantages:

1. High initial construction cost.
2. Environmental impact on aquatic life and local habitats.

3. Displacement of people and communities during dam construction.
4. Dependent on rainfall and water availability.

1.1.4 Diesel Engine Power Plant:

A **Diesel Power Plant** is a type of thermal power station where a **diesel engine** is used to drive an **electric generator (alternator)** to produce electricity. These plants are mainly used for **standby, emergency, or small-scale power generation**, especially in areas where grid power is unavailable.

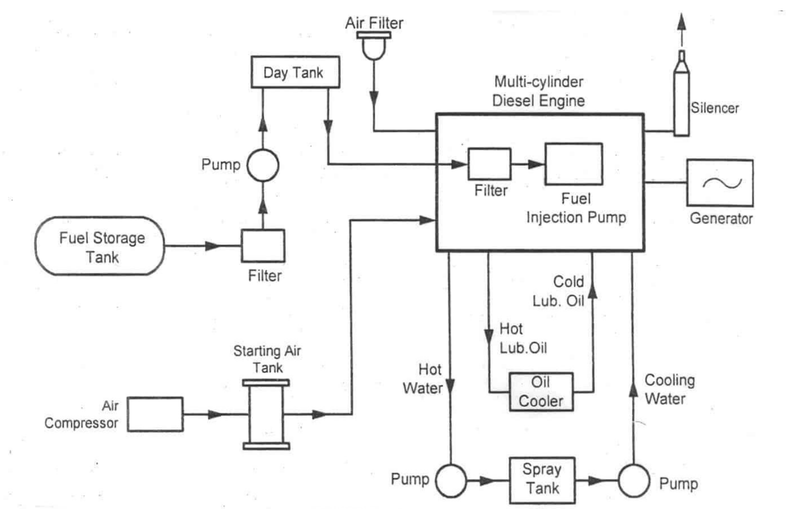

fig.3.4 Diesel Engine Power Plant

Working Principle:

The plant operates based on the **internal combustion (IC) engine principle**, where diesel fuel is burned inside the engine cylinder. The resulting high-pressure gases push the piston, rotating the crankshaft and turning the generator.

Main Components:

1. **Diesel Engine** – Burns diesel fuel to produce mechanical power.
2. **Air Intake System** – Supplies clean air to the engine.
3. **Fuel System** – Stores and supplies diesel to the engine.
4. **Cooling System** – Maintains engine temperature (usually water-cooled).
5. **Lubrication System** – Reduces friction and wear of moving parts.

6. **Exhaust System** – Removes combustion gases.
7. **Alternator (Generator)** – Converts mechanical energy into electrical energy.
8. **Control Panel** – Manages plant operation and monitors performance.

Advantages:

- Quick start-up and shutdown.
- Compact and easy to install.
- Useful for remote or off-grid areas.
- Lower capital cost compared to other power plants.

Disadvantages:

- High fuel cost and operational expenses.
- Emits pollutants (CO2, NO2, particulate matter).
- Not suitable for large-scale or continuous power supply.
- Requires regular maintenance.

1.2 Mechanical Transmission system:

Mechanical power transmission refers to the transfer of mechanical energy (physical motion) from one component to another in machines. Most machines need some form of mechanical power transmission. Common examples include electric shavers, water pumps, turbines and automobiles.

1.2.1 TYPES:

1. **Belt Drives**
2. **Rope drives**
3. **Chain drives**
4. **Gear drives**

1. Belt Drive system:

A **belt drive system** is a mechanical system used to transmit power between rotating shafts using a **belt** and **pulleys**. It is widely used in machinery, automotive systems, and industrial applications.

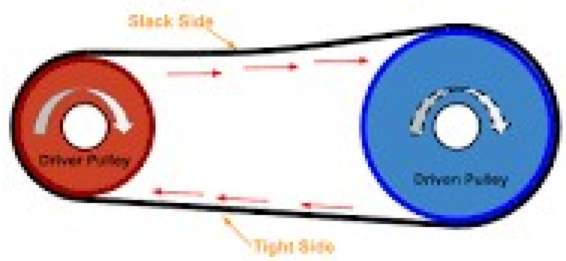

fig.3.5 Belt Drive

Main Components**:**

1. Driver Pulley **– Connected to the power source (e.g., motor).**
2. Driven Pulley **– Connected to the machine or load.**
3. Belt **– Flexible loop that transmits motion and power.**
4. Tight Side & Slack Side **– The belt has one side under tension (tight) and the other under less tension (slack).**

Advantages:

- Simple and cost-effective.
- Quiet and smooth operation.
- Shock absorber.
- Easy maintenance.

Disadvantages:

- Slippage can occur.
- Less efficient than gear drives.
- Belt wear and stretching over time.

Applications:

- The belt drive is used in the Mill industry.
- The belt drive is used in Conveyor.
- Power transmission in industrial machinery and equipment.
- Automotive applications, in different types of engines and transmissions.

- Agricultural machinery, such as tractors and harvesters.

2. Rope Drive :

A rope drive is a type of mechanical power transmission system that uses ropes instead of belts to transfer motion and power between rotating shafts, especially over long distances and where high power needs to be transmitted.

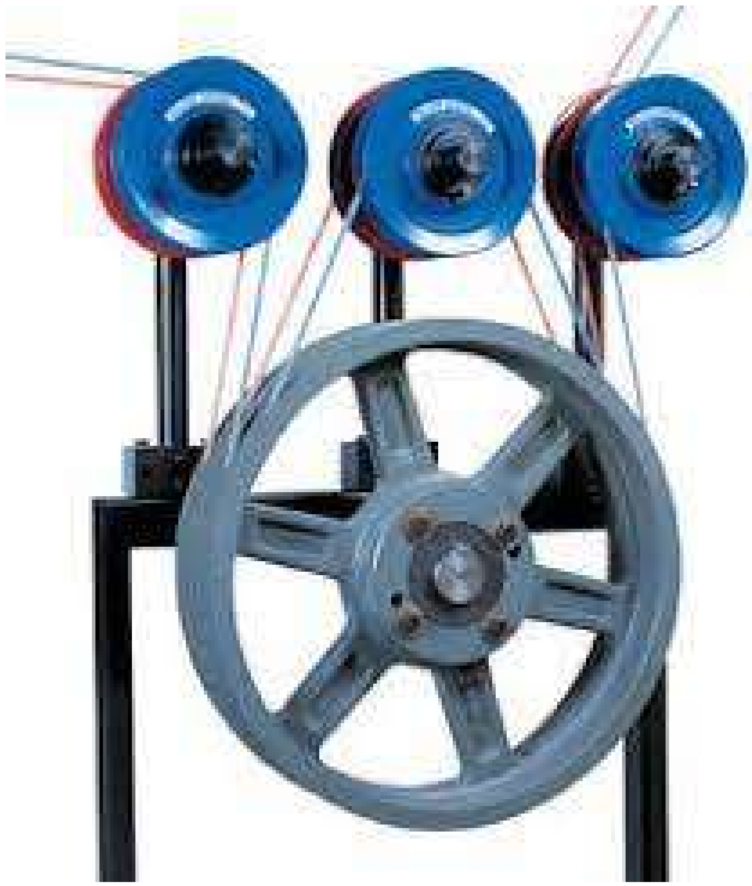

fig.3.6 Rope Drive

Main Components:

- **Rope**: Transmits motion and power.
- **Pulleys**: Have grooves to grip the rope and guide it.
- **Shafts**: Connect to pulleys; one is the driver, the other is the driven shaft.
- **Tensioning mechanism**: Ensures proper rope tension to prevent slippage.

Working Principle:

- The driver pulley rotates using a prime mover (like a motor).
- The rope transmits rotational motion to the driven pulley.
- The system works on the principle of **friction** between the rope and the pulley's grooves.

Advantages:

- Suitable for **long-distance** power transmission.
- **Flexible and silent** operation.
- Capable of transmitting **high power**.
- Can accommodate **slight misalignments** between shafts.

Limitations:

- Requires **regular maintenance** (lubrication, tensioning).
- Less efficient over **very long periods** due to stretch or wear.
- **Slip** can occur if tension is not maintained properly.

Applications:

- Overhead cranes
- Mining equipment
- Elevators
- Early factory machinery

3. Chain Drive :

A chain drive is a mechanical system that transmits power between two rotating shafts using a roller chain and sprockets.

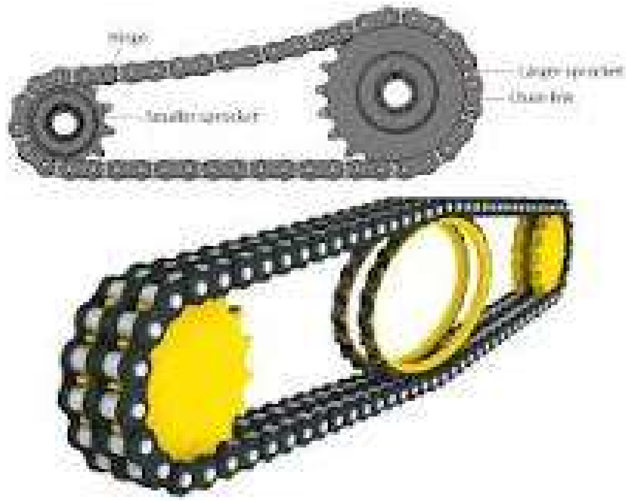

Fig 3.7 Chain Drive

Main Components:

- **Chain**: Made of links (typically roller chain).
- **Sprockets**: Toothed wheels that engage with the chain.
- **Shafts**: Driver and driven shafts connected to the sprockets.
- **Tensioner (optional)**: Maintains proper chain tension.

Working Principle:

- The driver sprocket rotates, pulling the chain.
- The chain moves in a loop, causing the driven sprocket to rotate.
- Motion is transferred with no slip due to positive engagement between the chain and sprockets.

Advantages:

- No slip – Positive drive.
- Can transmit high torque and power.
- Suitable for moderate distances.
- Durable and reliable under harsh conditions.
- Less maintenance than belt or rope drives (no stretching like ropes).

Limitations:

- Noisy compared to belt or rope drives.
- Requires lubrication and alignment.
- Cannot be used over very long distances.
- Prone to wear and chain elongation over time.

Applications:

- Bicycles and motorcycles
- Conveyors
- Agricultural machinery
- Industrial drives
- Timing mechanisms in engines

4. Gear Drive :

A gear drive is a power transmission mechanism that transfers motion and torque between shafts using meshing gear teeth.

fig.3.8 Gear Drive

- Gears – Toothed wheels that mesh together.
- Shafts – Connect the gears (driver and driven).
- Bearings – Support shafts and reduce friction.
- Housing – Encloses the gear assembly (optional).

Working Principle:

- When the driving gear rotates, its teeth engage with the driven gear, causing it to rotate.
- Motion is transferred without slip, and the direction of rotation depends on gear type.

Advantages:

- No slip, highly accurate speed ratio.
- Compact and capable of high power transmission.
- Suitable for short center distances.
- Can change speed, torque, and direction of rotation.

Limitations:

- Requires precision manufacturing.
- Costlier than belt, rope, or chain drives.
- Needs regular lubrication and alignment.
- Not suitable for long-distance transmission.

Applications:

1. Metal cutting
2. Power plants
3. Marine Engines
4. Clocks
5. Fuel pumps

1.3 Introduction to Robotics:

A robot is a machine especially one programmable by a computer capable of carrying out a complex series of actions automatically. A robot can be guided by an external control device, or the control may be embedded within.

Fig. 3.9 Robot

1.3.1 Robot Joints & Links:

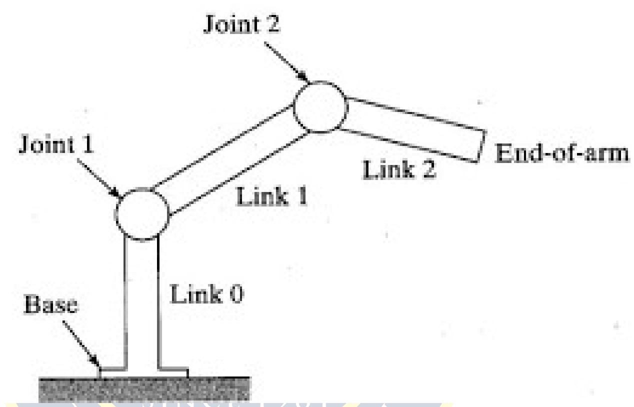

fig. 3.10 Robot Joints & Links

Links:

- Links are the rigid bodies in a robot.
- They form the arms or body segments of the robot.
- Connect joints together.
- Do not move or bend — they just transfer motion.

Joints:

- **Joints** (also called **actuators**) allow **relative motion** between two links.
- They control the **movement** of the robot.

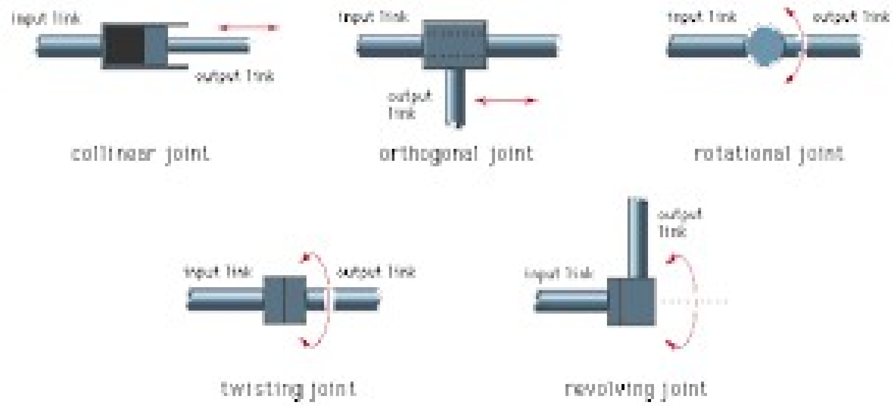

fig. 3.11 Robot Joints & Links

The links are the rigid members connecting the joints. The joints (also called axes) are the movable components of the robot that cause relative motion between adjacent links.

Robot Configurations:

The configuration of a robot is a specification of the position of all points of the robot, and the space of all configurations of the robot is called the configuration space or C-space of the robot.

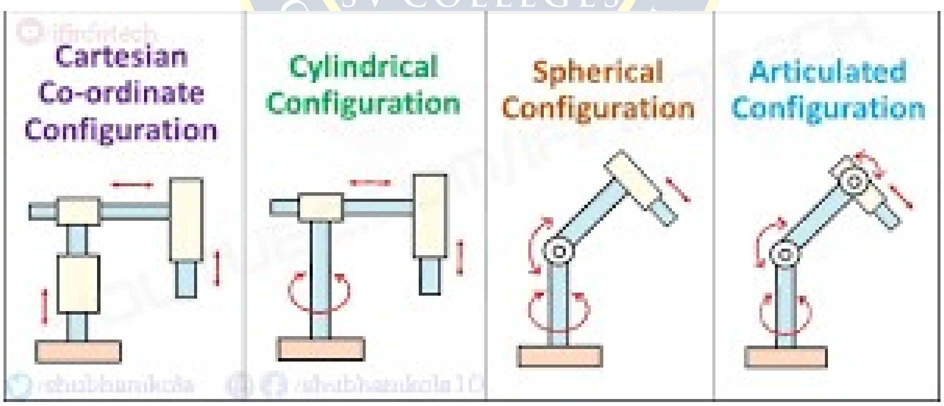

fig.3.12 Robot Configurations

The configuration of a robot is a specification of the position of all points of the robot, and the space of all configurations of the robot is called the configuration space or C-space of the robot.

Robot Applications:

1. Manufacturing
2. Security
3. Space Exploration
4. Entertainment
5. Agriculture
6. Health Care
7. Underwater Exploration
8. Food Preparation
9. Military
10. Customer Service

9. Practice Quiz

1. What is the primary fuel in a thermal power plant?
Thermal power plants use coal, oil, or gas to heat water and produce steam for electricity.

2. How does a nuclear power plant produce energy?
Energy is produced through nuclear fission, where uranium atoms split and release heat.

3. What is the energy source for hydro power plants?
Hydro power plants use the kinetic energy of flowing or falling water to drive turbines.

4. What engine type powers a diesel power plant?
A diesel power plant uses a compression ignition diesel engine to generate electricity.

5. What is a major drawback of nuclear power?
Nuclear power generates radioactive waste that is hazardous and difficult to dispose of

6. What is a belt drive?
A flexible belt transmits power between pulleys; used in low power, smooth, and silent drives.

7. What is a rope drive?
Ropes transmit power between pulleys over long distances with moderate load capacity.

8. What is a chain drive?
Chains and sprockets transmit power efficiently with no slip; used in bikes and conveyors.

9. What is a robot?
A robot is a programmable machine that performs tasks automatically or with guidance.

10. What are robots used for in space?

Used for satellite servicing, exploration, and data collection in space missions.

10. Assignments

S.No	Question	BL	CO
1	Briefly explain the elements and working of Hydro Electric Power plant with neat sketch.	2	3
2	Differentiate between Belt drive, Chain drive and Rope drive with examples?circuit	3	3
3	Give complete note on working principle of a Thermal power plant with neat diagram.	1	3
4	Describe about Robot configurations with neat sketches?	4	3

11. Part A- Question & Answers

S.No	Question & Answers	BL	CO
1	**What are the drives used in mechanical power transmission** **Belt Drive** – Uses belts and pulleys; suitable for moderate power and long-distance transmission. **Rope Drive** – Uses ropes for transmitting power over long distances; handles heavy loads. **Chain Drive** – Uses chains and sprockets; provides positive, slip-free transmission. **Gear Drive** – Uses interlocking gears; offers precise and compact power transfer.	1	3
2	**What is Turbine?** A turbine is a rotary device that converts the energy of a moving fluid (like steam, water, or gas) into mechanical energy, which is often used to generate electricity.	1	3
3	**What are the advantages of thermal power plant over nuclear power plant?** **Lower initial cost** to build compared to nuclear plants. **Easier fuel availability** (coal, gas) versus limited uranium supply. **Simpler technology** and operation.	1	3
4	**Write the applications of belt drive?** 1. Agricultural machinery like threshers and harvesters. 2. Conveyors in manufacturing and packaging industries.	1	3

	3. Fans and blowers for air circulation. 4. Machine tools such as lathes and drills.		
5	**Write the applications of Rope drive?** 1. Elevators and hoists for lifting heavy loads. 2. Cable railways and ropeways for transportation. 3. Long-distance power transmission where belts are not suitable.	1	3
6	**Write the applications of gear drive?** 1. Automobile gearboxes for speed and torque control. 2. Machine tools like lathes and milling machines. 3. Clocks and watches for precise movement control. 4. Conveyor systems in industries.	1	3
7	**Define the term Robot** A robot is a programmable machine designed to perform tasks automatically, either autonomously or by remote control.	1	3
8	**Write the applications of chain drive?** 1. Bicycles and motorcycles for power transmission to wheels. 2. Conveyor systems in manufacturing and packaging. 3. Automobile engines for timing and camshaft drives. 4. Agricultural machinery like harvesters and tractors.	1	3
9	**What are the advantages of hydro power plant over thermal power plant?** 1. Renewable energy source using water flow. 2. No fuel cost as it uses natural water. 3. Clean and eco-friendly with no air pollution. 4. Low operating and maintenance cost.	1	3
10	**Write the applications of Robots?** 1. Industrial automation: Welding, painting, assembly, and material handling. 2. Medical field: Surgery assistance, rehabilitation, and patient care. 3. Space exploration: Satellite servicing and planetary exploration. 4. Military: Bomb disposal, surveillance, and reconnaissance. 5. Agriculture: Planting, harvesting, and crop monitoring.	1	3

12. Part B- Questions

S.No	Question	BL	CO
1	Explain working principle of Steam Power plant with neat sketch. State it's advantages and disadvantages	1	3
2	Discuss in detail working principle of Nuclear Power plant with neat sketch.	2	3
3	Briefly explain the elements and working of Hydro Electric Power plant with neat sketch.	3	3
4	Explain with neat sketch the working principle of Diesel Engine Power plant. write its advantages and disadvantages.	2	3
5	Differentiate between Belt drive, Chain drive, Gear drive and Rope drive with examples?	3	3
6	Define the term Robot and explain the different links and joints used in robots with neat sketches. Also explain configurations of Robot?	2	3

13. Supportive Online Certification Courses

1. Thermal power plants By Prof. Ravi Kumar, IIT Roorkee – 8 weeks
2. Robotics: Advanced Concepts By Prof. C.S. Shankar Ram, IIT Madras– 12 weeks.

14. Real Time Applications

S.No	Application	CO
1	Power plants are the primary source of electricity for homes, offices, schools, hospitals, and commercial buildings. They operate continuously to ensure an uninterrupted power supply to cities and towns, meeting day-to-day energy demands.	3
2	In tractors, harvesters, and other farm equipment, mechanical transmissions like gear drives and belt drives are used to transfer power from the engine to tools such as plows or cutters. They ensure reliability in rough terrains and heavy-duty operations.	3
3	Chain and belt drives are widely used in manufacturing industries to transmit power between machines and conveyor belts. These systems	3

	allow for efficient, continuous operation in production lines, reducing downtime and increasing output.	
4	Robots are widely used in manufacturing industries for tasks like welding, painting, assembly, and material handling. They improve precision, speed, and consistency, reducing human error and increasing production efficiency. Automotive factories are a prime example.	**3**

15. Contents Beyond the Syllabus

1. Advanced and emerging topics in power plants

16. Prescribed Text Books & Reference Books

Text Books:

1. R.S. Khurmi & J.K. Gupta – "A Textbook of Basic Mechanical Engineering" – S. Chand Publications
2. Sharma & Purohit – "Basic Mechanical Engineering" – Pearson

References:

1. P.K. Nag – "Engineering Thermodynamics" – Tata McGraw Hill
2. P.L. Ballaney – "Thermal Engineering" – Khanna Publishers

17. Mini Project Suggestion

1. **Thermal Power Plant Working Model:**
   Create a demo using heaters, a steam setup, and a small turbine to simulate electricity generation.
2. **Gear Train Mechanism Demo:**
   Design a model showing gear ratios and speed variation using spur, helical, and bevel gears.
3. **Rope Pulley Transmission Model:**
   Develop a model to demonstrate mechanical advantage and load lifting with different pulley systems.
4. **Pick and Place Robotic Arm:**
   Build a robotic arm using servos to pick small objects and place them using programmed control.